

$$\backslash \}$$

$$T=0.7$$

$$\ell i \ell$$

$$y_\ell \in \{+1,-1\}$$

$$x_{\ell,i} \in \{0,1\} i \ell$$

$$im_iu_i$$

$$\begin{aligned}\mu_\ell &= \sum_i x_{\ell,i} m_i, \\ \tau_\ell &= \sqrt{\sum_i x_{\ell,i} \exp(2u_i) + \varepsilon}, \\ z_\ell &= y_\ell \mu_\ell / \tau_\ell.\end{aligned}$$

$$p(y_\ell \mid \{m_i,u_i\}) = \Phi(z_\ell),$$

$$\Phi$$

$$\mathcal{J} = -\sum_{\ell} \log \Phi(z_{\ell}) + \frac{1}{2\sigma_m^2} \sum_i m_i^2 + \frac{1}{2\sigma_u^2} \sum_i u_i^2.$$

$$\sigma_m=1.0\sigma_u=1.0\varepsilon=10^{-6}[-4,2]\log\Phi$$

$$\backslash \}$$

$$r_i = \frac{m_i}{\exp(u_i) + \varepsilon},$$

$$\tilde{r}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r) + \varepsilon},$$

$$s_i = \exp(-\alpha \tilde{r}_i^2), \alpha = 1.$$

$$s_p s_{c_1}, \ldots, s_{c_b}$$

$$s=\frac{bs_p+\sum_{j=1}^bs_{c_j}}{2b}.$$

$$k$$



$$0.1$$

$$\gamma(b)=\exp\big(-k(b-1)\big),$$

$$k=0.35$$

$$m_i/\exp(u_i)$$

$$\mathcal{L} = \frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} \text{clip}(A_t, -c, c) \cdot \text{CE}\left(p_{\theta}(\cdot \mid x_{<t}), x_t\right),$$

$$\mathcal{S}A_t[-3,3]$$

$$0.5$$

$$0.7$$
